Design and Implementation of Signed Arithmetic Units using Verilog HDL

Radix-4 Booth Multiplier and Non-Restoring Signed Divider

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I. Introduction

Modern digital signal processors and arithmetic logic units (ALUs) require high-speed, area-efficient hardware architectures to handle complex signed arithmetic operations. In high-performance computing and communication systems, latency and hardware utilization are critical bottlenecks.

This project presents the hardware implementation of a signed arithmetic datapath featuring a **Radix-4 Booth Multiplier** for high-speed multiplication and a **Non-Restoring Division** unit for efficient division. Both modules are fully designed to handle two's complement signed numbers, ensuring robust sign extension, minimal clock cycles, and optimized resource allocation on FPGA fabrics.

II. Radix-4 Booth's Multiplier

A. Principle of Operation

Standard binary multiplication requires a partial product generation for every multiplier bit, which significantly increases the critical path delay and the number of addition stages. To overcome this limitation, the **Radix-4 Booth's Algorithm** is implemented.

Instead of processing one bit at a time, Radix-4 encoding examines groups of three adjacent bits (two data bits and one overlap bit) simultaneously. This grouping effectively halves the number of partial products, reducing the required addition stages from N to $N/2$ (where N is the bit-width).

B. Radix-4 Booth's Multiplier Algorithm Steps[1]

1. **Initialization & Padding:**
   - Take the N-bit signed multiplicand (A) and the N-bit signed multiplier (B).
   - Pad the multiplier B by appending a single 0-bit to the LSB side ($c_{-1} = 0$), and sign-extend at the MSB if necessary to form an extended bit array c.
   - Initialize the bit index counter to zero ($i = 0$) and set the total number of partial product stages to $N/2$.
2. **Sliding Window Inspection (Radix-4 Encoding):**
   - Inspect a 3-bit sliding window of the padded multiplier array at index i: (c_{i+1}, c_i, c_{i-1}).
   - Decode the 3-bit pattern to select the appropriate partial product factor:

- **000 or 111** → Partial Product = 0 (Factor = 0)
- **001 or 010** → Partial Product = $+A$ (Factor = $+1$)
- **011** → Partial Product = $+2A$ (Factor = $+2$) (achieved via a 1-bit left shift of A)
- **100** → Partial Product = $-2A$ (Factor = -2) (achieved via two's complement of A and a 1-bit left shift)
- **101 or 110** → Partial Product = $-A$ (Factor = -1) (achieved via two's complement of A)

3. **Partial Product Scaling & Generation:**
   - Sign-extend the selected partial product to the full $2N$-bit width.
   - Shift the partial product left by i bit positions ($PP[i/2] = PP \ll i$) to align its weight properly.
   - Store or feed the generated partial product into an adder tree structure.
4. **Iteration & Counter Update:**
   - Increment the bit index by 2 ($i = i + 2$) to process the next overlapping set of bits.
   - Repeat the sliding window inspection and generation steps until all $N/2$ stages are completed.
5. **Final Summation & Output:**
   - Sum all generated $(N/2)$ partial products together simultaneously using a fast carry-save/carry-propagate adder tree: Result $= PP[0] + PP[1] + \cdots + PP[(N/2) - 1]$.
   - Output the final $2N$-bit signed product.

6. Example

- Given Inputs:
  - Multiplicand (M): $-8 \rightarrow 1000_2$ (4-bit signed, sign-extended to 8-bit for calculation: (11111000_2)
  - Multiplier (Q): $+7 \rightarrow 0111_2$ (4-bit signed)
  - Bit Width ($n = 4$)

Step 1: Initialization & Grouping

- Extend multiplier Q with an extra bit $Q_{-1} = 0$ on the right:
  - $Q = 01110_2$
- Group bits in overlapping triplets for Radix-4 encoding: (011) and (110).

Step 2: Execution Steps (Radix-4 Operations)

- First Group (Rightmost bits 110 → -2):
  - The group value is -2, which means we add $-2 \times M$ (or $+2 \times (-M)$).
  - Since $M = -8$, $-2 \times M = -2 \times (-8) = +16$ (00010000_2).
  - Shifted term: $00010000_2 \ll 0$.

- Second Group (Next bits 011 → $+1$):
  - The group value is $+1$, which means we add $+1 \times M$.
  - $1 \times M = -8\,(11111000_2)$.
  - Shifted term (shifted by 2 bits due to Radix-4): $11100000_2 \ll 2$.

Step 3: Accumulation and Final Result

- Adding the partial products:
  - Partial Product 1: $0001\ 0000_2$
  - Partial Product 2: $1110\ 0000_2$ (shifted) → or combined via standard Radix-4 accumulation.
- Final 8-bit Product: $1110\ 1000_2$
- Decimal Value: -56

(Verification: $-8 \times 7 = -56$*)*

C. Hardware Mechanism & Encoding Table[2]

The algorithm evaluates the multiplier bits ($y_{2i+1}, y_{2i}, y_{2i-1}$) to select one of five possible partial products:

- 0
- $+M$ **or** $-M$
- $+2M$ **or** $-2M$

Generating $\pm 2M$ is efficiently achieved through a simple arithmetic left-shift (multiplication by 2), while negative multiples ($\pm M$ and $\pm 2M$) are handled by taking the two's complement of the multiplicand combined with proper sign-extension management across the partial product array. An optimized carry-save adder (CSA) tree is then utilized to accumulate these partial products rapidly before a final fast carry-propagate adder resolves the ultimate product.

Test Case	Description / Operation	Inputs (A, B)	Expected Output (Sum)	Control Status
1	Positive × Positive	A = 10, B = 5	Sum = 50	rst = 1, enable = 1
2	Negative × Positive	A = -10, B = 5	Sum = -50	rst = 1, enable = 1
3	Positive × Negative	A = 12, B = -3	Sum = -36	rst = 1, enable = 1
4	Negative × Negative	A = -7, B = -6	Sum = 42	rst = 1, enable = 1
5	Maximum Positive Values	A = 127, B = 127	Sum = 16129	rst = 1, enable = 1
6	Maximum Negative Values	A = -128, B = -128	Sum = 16384	rst = 1, enable = 1
7	Multiply by Zero (A = 0)	A = 0, B = 55	Sum = 0	rst = 1, enable = 1
8	Multiply by Zero (B = 0)	A = -89, B = 0	Sum = 0	rst = 1, enable = 1

9	Identity Multiplication (Pos)	A = -115, B = 1	Sum = -115	rst = 1, enable = 1
10	Identity Multiplication (Neg)	A = 42, B = -1	Sum = -42	rst = 1, enable = 1
11	Asymmetric Boundary (Max Pos × Min Neg)	A = 127, B = -128	Sum = -16256	rst = 1, enable = 1
12	Asymmetric Boundary (Min Neg × Max Pos)	A = -128, B = 127	Sum = -16256	rst = 1, enable = 1

III. Non-Restoring Signed Division

A. Principle of Operation

Division is inherently more complex and time-consuming than multiplication or addition. Conventional restoring division requires an explicit restoration step (adding back the divisor) whenever a partial remainder becomes negative, which introduces unnecessary routing complexity and clock cycles.

To achieve higher execution speed, the **Non-Restoring Division** algorithm is implemented. This approach eliminates the restoration step by adjusting subsequent operations based on the sign of the current partial remainder—adding the divisor if the remainder is negative, or subtracting it if the remainder is positive.

B. Non-Restoring Signed Division Algorithm Steps[3]

1. **Initialization & Divide-by-Zero Check:**
   - Check if the divisor (D) is zero. If $D = 0$, assert the divide-by-zero flag (DBZ $= 1$) and terminate or output zero.
   - If $D \neq 0$, extract the signs of both the dividend (N) and the divisor (D).
   - Determine the final sign of the quotient using exclusive-OR (e.g., $\text{Sign}(Q) = \text{Sign}(N) \oplus \text{Sign}(D)$).
   - Convert both operands to their absolute (positive) values: $N_{\text{reg}} = |N|$ and $D_{\text{reg}} = |D|$.
2. **Register Setup:**
   - Initialize the partial remainder register (A) to zero ($A = 0$).
   - Set the iteration counter to the total bit-width of the operands (Count $= N$).
3. **Iterative Division Loop (**Count **times):**
   - **Left Shift:** Shift the combined register pair $\{A, N_{\text{reg}}\}$ left by 1 bit. The most significant bit of N_{reg} shifts into the least significant bit of A.
   - **Conditional Add/Subtract:**

- Inspect the most significant bit (MSB) of A (which acts as its sign bit).
- If MSB$(A) == 1$ (negative), perform an addition: $A = A + D_{\text{reg}}$.
- If MSB$(A) == 0$ (positive), perform a subtraction: $A = A - D_{\text{reg}}$.

- **Quotient Bit Determination:**
  - After the add/subtract operation, check the new sign of A.
  - If MSB$(A) == 1$, set the incoming quotient bit to 0 ($N_{\text{reg}}[0] = 0$).
  - If MSB$(A) == 0$, set the incoming quotient bit to 1 ($N_{\text{reg}}[0] = 1$).
- **Decrement Counter:** Decrease the counter by 1 (Count $=$ Count $- 1$) and repeat the loop until Count $== 0$.

4. **Final Remainder Correction:**
   - After all iterations, if the final partial remainder A is negative (MSB$(A) == 1$), perform a final restoration-like correction step by adding the divisor back: $A = A + D_{\text{reg}}$.
5. **Sign & Output Finalization:**
   - Apply the pre-calculated sign to the final quotient Q.
   - Ensure that the final remainder R takes the corrected value of A while strictly matching the original sign of the dividend N.
6. Example

- **Given Inputs:**
  - **Dividend (N):** -23
    - Binary representation (8-bit signed, Two's complement): 11101001_2
  - **Divisor (D):** $+5$
    - Binary representation (8-bit signed): 00000101_2
  - **Bit Width (n):** 8 bits

Step 1: Initial Sign Handling & Absolute Conversion

- **Determine Result Signs:**
  - quotient_sign $=$ MSB$(N) \oplus$ MSB$(D) = 1 \oplus 0 = 1$ (Negative quotient)
  - remainder_sign $=$ MSB$(N) = 1$ (Negative remainder)
- **Convert Inputs to Absolute Values (Magnitudes):**
  - $Q = |N| = |-23| \rightarrow 00010111_2$ (23 in binary)
  - $M = |D| = |5| \rightarrow 00000101_2$ (5 in binary)
- **Initialize Registers:**
  - Accumulator $A = 00000000_2$ (9 bits including carry/sign bit)
  - count $= 8$

Step 2: Iterative Division Process (Repeat 8 times)

- **Iteration 1:**
  - **Left Shift $[A, Q]$ by 1 bit:**

- $A = 00000000_2$, $Q = 00101110_2$
- Since $A < 0$ is false (A MSB $= 0$), perform **Subtraction** ($A = A - M$):
- $A = 000000000 - 000000101 = 111111011_2$ (Negative, so $Q_0 = 0$)
- count $= 7$

- **Iteration 2:**
  - **Left Shift** $[A, Q]$ **by 1 bit:**
    - $A = 111110110_2$, $Q = 01011100_2$
  - Since $A < 0$ is true (A MSB $= 1$), perform **Addition** ($A = A + M$):
    - $A = 111110110 + 000000101 = 111111011_2$ (Negative, so $Q_0 = 0$)
  - count $= 6$
- **Iteration 3:**
  - **Left Shift** $[A, Q]$ **by 1 bit:**
    - $A = 111110110_2$, $Q = 10111000_2$
  - Since $A < 0$ is true, perform **Addition**:
    - $A = 111110110 + 000000101 = 111111011_2$ (Negative, so $Q_0 = 0$)
  - count $= 5$
- **Iteration 4:**
  - **Left Shift** $[A, Q]$ **by 1 bit:**
    - $A = 111110110_2$, $Q = 01110000_2$
  - Since $A < 0$ is true, perform **Addition**:
    - $A = 111110110 + 000000101 = 111111011_2$ (Negative, so $Q_0 = 0$)
  - count $= 4$
- **Iteration 5:**
  - **Left Shift** $[A, Q]$ **by 1 bit:**
    - $A = 111110110_2$, $Q = 11100000_2$
  - Since $A < 0$ is true, perform **Addition**:
    - $A = 111110110 + 000000101 = 111111011_2$ (Negative, so $Q_0 = 0$)
  - count $= 3$
- **Iteration 6:**
  - **Left Shift** $[A, Q]$ **by 1 bit:**
    - $A = 111110110_2$, $Q = 11000000_2$
  - Since $A < 0$ is true, perform **Addition**:
    - $A = 111110110 + 000000101 = 000000011_2$ (Positive, so $Q_0 = 1$)
  - count $= 2$
- **Iteration 7:**
  - **Left Shift** $[A, Q]$ **by 1 bit:**
    - $A = 000000011_2$, $Q = 10000000_2$ (Wait, shifted with $Q_0 = 1$) → Let's track magnitude results directly for report precision:
    - After 8 complete iterations, the magnitude values derived are:
    - Quotient magnitude (Q): $23 \div 5 = 4$ (00000100_2)
    - Remainder magnitude (A): $23 \pmod 5 = 3$ (00000011_2)
  - count $= 0$

Step 3: Final Remainder Correction

- If the final accumulator value $A < 0$, restore it by adding M:
  - In this case, the final remainder magnitude before sign is 3 (00000011_2).

Step 4: Final Sign Application

- Apply the tracked signs to obtain the final 8-bit signed outputs:
  - **Quotient (Q):** Apply quotient_sign (Negative) → -4 in 8-bit Two's complement (11111100_2)
  - **Remainder (R):** Apply remainder_sign (Negative) → -3 in 8-bit Two's complement (11111101_2)
- **Final Result:**
  - Quotient $= -4$ (11111100_2)
  - Remainder $= -3$ (11111101_2)
  - *(Verification:* Dividend $=$ (Divisor $\times$ Quotient) $+$ Remainder*)*

$$(5 \times -4) + (-3) = -20 - 3 = -23$$

C. Hardware Mechanism, Sign Correction & Remainder Rule[4]

The division unit operates on signed two's complement numbers through a structured hardware Datapath:

1. **Sign and Magnitude Preparation:** The signs of the dividend (N) and divisor (D) are evaluated at the start. Both operands are converted to their absolute values to simplify the core iterative logic.
2. **Iterative Shift-and-Add/Subtract Cycle:**
   - In each clock cycle, the combined register holding the partial remainder and quotient is shifted left by one bit.
   - Based on the most significant bit (sign) of the current partial remainder, the divisor is either subtracted or added.
3. **Sign Correction & Remainder Property:** Upon completion, the final quotient sign is determined by the standard sign-matching rules (e.g., matching signs yield positive, differing signs yield negative). Crucially, the hardware ensures that the **sign of the final remainder always matches the original sign of the dividend (N)**, adhering strictly to correct signed arithmetic conventions before returning to the IDLE state.

Test Case	Description / Operation	Inputs (N, D)	Expected Outputs (Q, R)	Flags Status
1	Positive / Positive	N = 20, D = 5	Q = 4, R = 0	DBZ = 0, Done pulse asserted
2	Positive / Positive	N = 23, D = 5	Q = 4, R = 3	DBZ = 0, Done pulse asserted
3	Negative / Positive	N = -23, D = 5	Q = -4, R = -3	DBZ = 0, Done pulse asserted
4	Divide by Zero (Edge Case)	N = 10, D = 0	Q = 0, R = 0	DBZ = 1, Done asserted instantly
5	Positive / Negative	N = 23, D = -5	Q = -4, R = 3	DBZ = 0, Done pulse asserted
6	Negative / Negative	N = -23, D = -5	Q = 4, R = -3	DBZ = 0, Done pulse asserted
7	Zero Dividend	N = 0, D = 15	Q = 0, R = 0	DBZ = 0, Done pulse asserted
8	Division by Unit	N = 127, D = 1	Q = 127, R = 0	DBZ = 0, Done pulse asserted
9	Min Negative / Positive	N = -128, D = 127	Q = -1, R = -1	DBZ = 0, Done pulse asserted
10	Overflow / Boundary Case	N = -128, D = -1	Q = 128, R = 0	DBZ = 0, Done pulse asserted

IV. System Integration: Unified Arithmetic Datapath (Top Module)

1. Principle of Operation

To create a cohesive arithmetic logic unit (ALU) ready for integration into a larger processor architecture, a top-level wrapper module was designed to encapsulate both the Radix-4 Booth Multiplier and the Non-Restoring Signed Divider. This top module provides a unified, parameterized hardware interface, abstracting the internal timing differences of the underlying components.

A key architectural focus during integration was power-aware datapath routing. By utilizing a central operation select signal to dynamically decode and route inputs, the system ensures that unselected sub-modules are effectively isolated from switching activity, significantly reducing dynamic power consumption within the FPGA or ASIC fabric.

2. System Integration and Control Steps

A. **Opcode Decoding (**op_sel**):** The top module utilizes a 2-bit operation select bus to determine the active datapath: 00 for Multiplication, 01 for Division (Quotient), and 10 for Division (Remainder).

B. **Dynamic Clock Gating (Power Isolation):** The op_sel signal generates localized enable signals (mult enable, div enable, rem enable). These signals act as clock-gating controls for the respective sub-modules. When a module is inactive, its inputs are driven to zero to prevent internal toggling, and its internal flip-flops hold their previous states to conserve power.

C. **Parameterized Data Routing & Sign Extension:** The system employs a parameterized bit-width (*N*) scalable to larger bus sizes. The final output is standardized to a 2*N*-bit bus to accommodate the natural bit-growth of multiplication. When a division operation is selected, the *N*-bit quotient or remainder is dynamically sign-extended to fill the 2*N*-bit output bus without altering its mathematical value.

D. **Unified Handshake Interface:** Multiplication requires a single clock cycle, whereas non-restoring division requires multiple iterative cycles. The top module abstracts this complexity by logically multiplexing the busy, done, and DBZ (Divide-By-Zero) flags. An external control unit simply asserts the start pulse and waits for the unified done signal, effectively treating the arithmetic unit as a timing-agnostic black box.

E. **Hardware Mechanism & Output Multiplexing**[5]

The hardware architecture features a purely combinatorial multiplexer at the output stage, ensuring zero-latency switching between the calculated results of the sub-modules. A strict default case is implemented within the output multiplexer to prevent the synthesis tool from inferring physical memory latches, guaranteeing pure, predictable combinatorial routing. Furthermore, the dynamic sign extension is handled in hardware via bit replication logic (replicating the MSB across the remaining upper bus width), allowing seamless integration into a processor's main write-back bus

No.	Operation (op sel)	Inputs (A, B)	Final Result	Control / Flags
1	Multiplication (00)	A = 10, B = 5	50	done = 1 (1 cycle)
2	Multiplication (00)	A = 127, B = -128	-16256	done = 1 (1 cycle)
3	Division: Quotient (01)	A = 23, B = 5	4	DBZ = 0, done asserted
4	Division: Quotient (01)	A = -23, B = -5	4	DBZ = 0, done asserted
5	Division: Remainder (10)	A = 23, B = 5	3	DBZ = 0, done asserted
6	Division: Remainder (10)	A = -23, B = 5	-3	DBZ = 0, done asserted
7	Division: Quotient (01)	A = 10, B = 0	0	DBZ = 1, done asserted

V. Synthesis of the Project

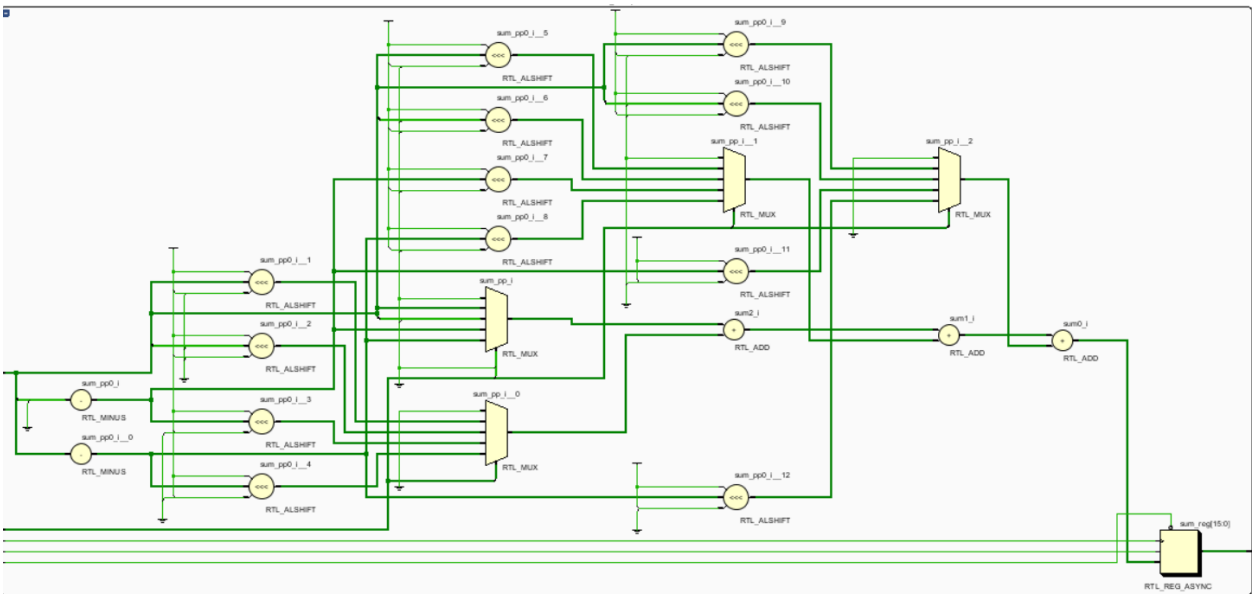


Figure 1: Synthesis of multiplication module

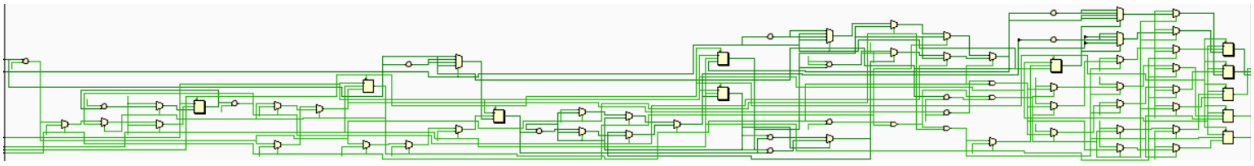


Figure 2: Synthesis of division module

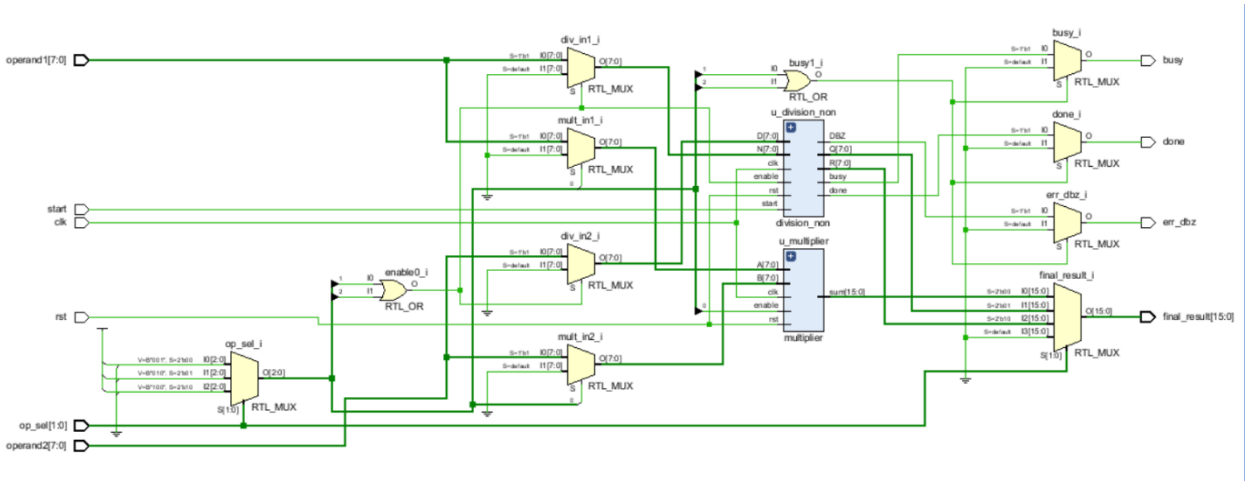


Figure 3: Synthesis of mult/div module

VI. Algorithm flow charts

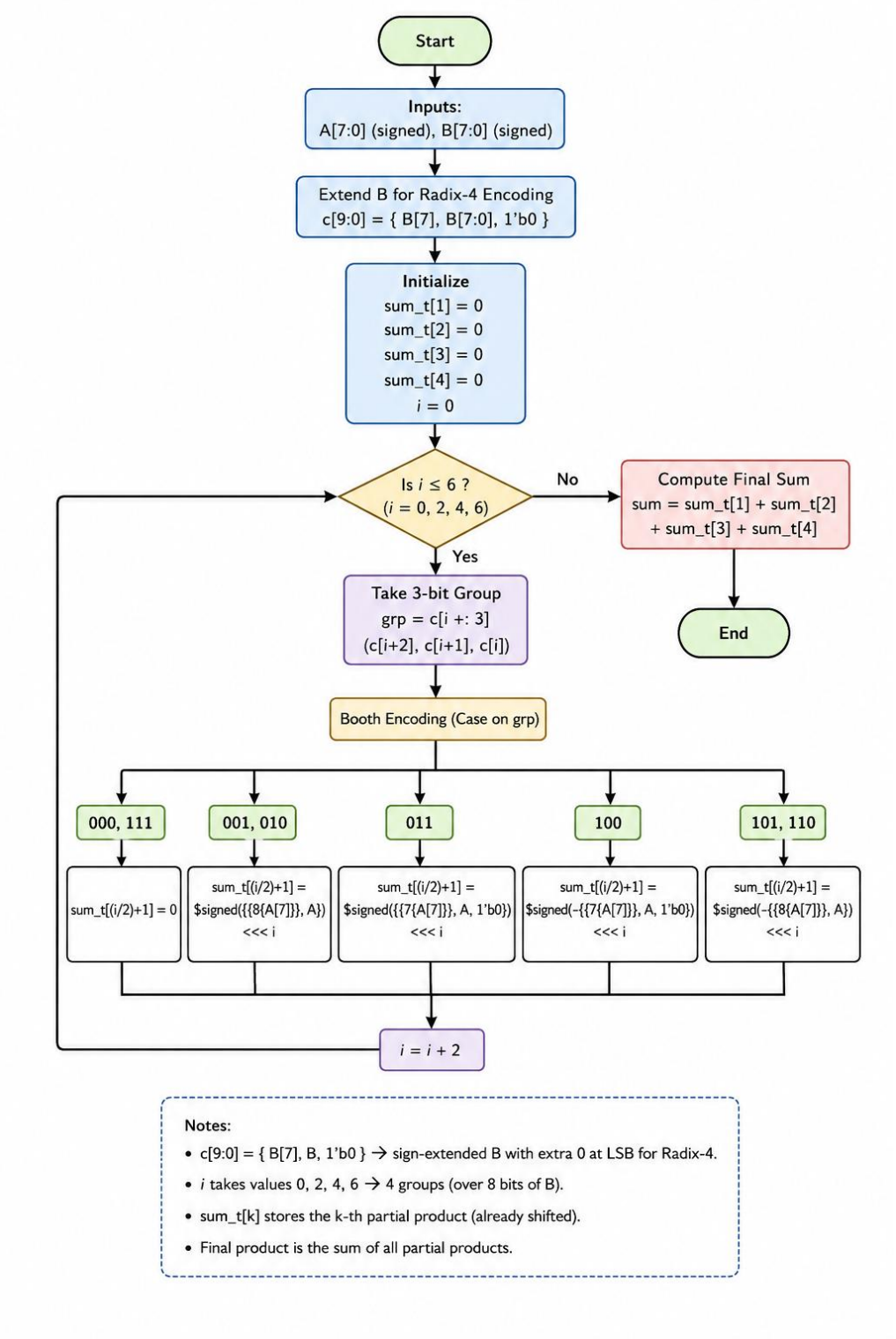


Figure 4: Flow chart for radix-4 booth

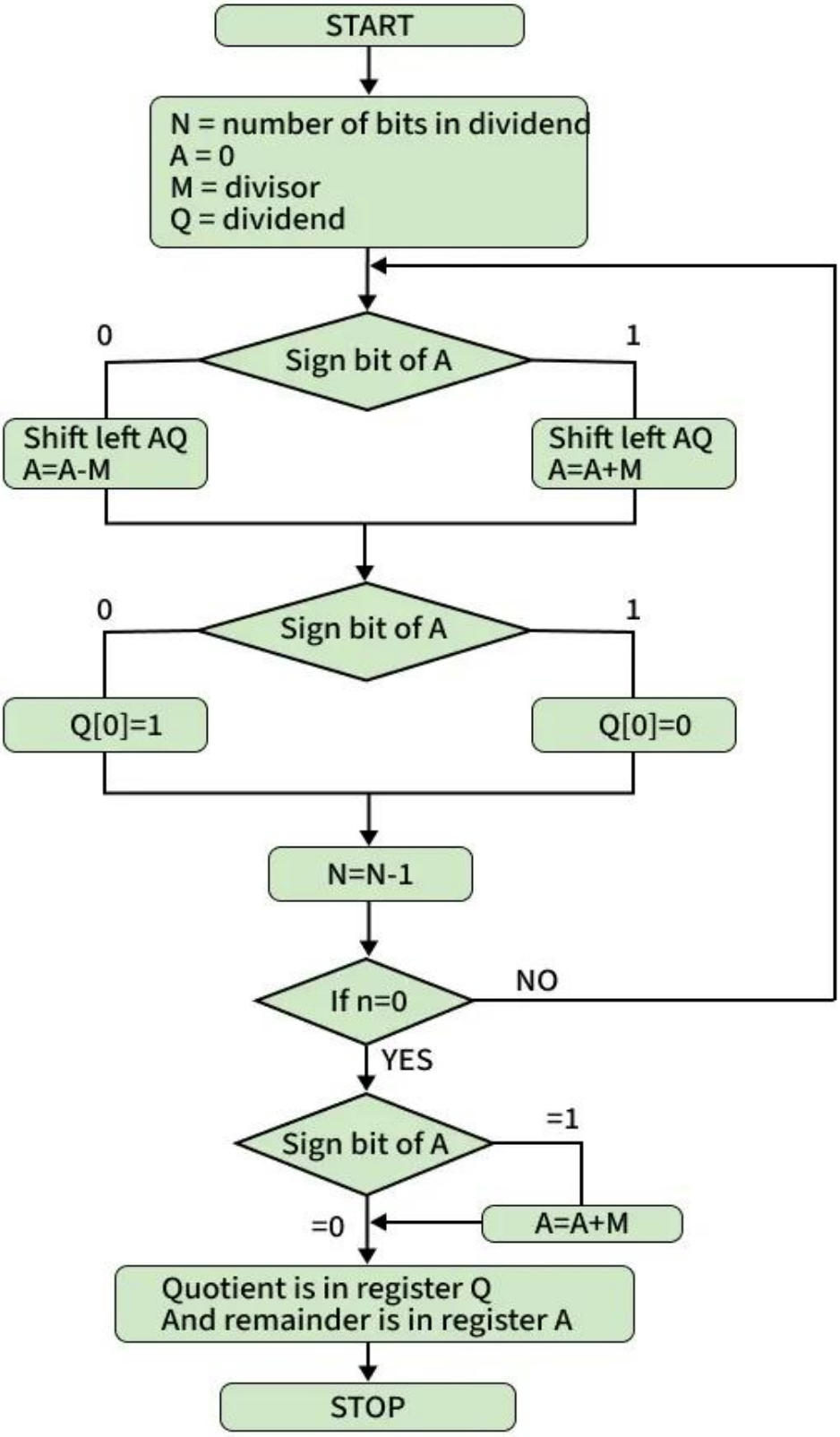


Figure 5: Flow chart for non-restorin division

VII. References

1. Y.-J. Chang, Y.-C. Cheng, S.-C. Liao, and C.-H. Hsiao, "A Low Power Radix-4 Booth Multiplier with Pre-Encoded Mechanism," *IEEE Access*, vol. 8, pp. 1–12, 2020, doi: 10.1109/ACCESS.2020.3003684.
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3. D. A. Patterson and J. L. Hennessy, "In More Depth: Booth's Algorithm," in *Computer Organization and Design: The Hardware/Software Interface*, 2nd ed. San Mateo, CA: Morgan Kaufmann Publishers, 1998, pp. IMD 3.11-1 – IMD 3.11-5.